PRODUCTSCOPE

SaaS Product Performance & Customer Analytics

Turning Customer Behavior into Actionable Business Insights

Prepared by	Gaurav Kumar
Profile	Computer Science Student | Aspiring Data Analyst
Tools	Python, Pandas, SQL, Jupyter Notebook, Power BI
Scope	Acquisition, activation, conversion, feature adoption, churn and subscription revenue

A practical analytics case study connecting product usage behavior with customer conversion and subscription outcomes.

1. Executive Summary

ProductScope analyzes SaaS product data across users, product events, sessions and subscriptions. It combines Python-based exploration, SQL business analysis and Power BI reporting to examine the journey from signup through activation, trial and paid subscription.

Headline findings: the supplied processed outputs report 10,000 users, 6,314 activated users (63.14%), 3,097 trial users and 1,972 paid users. Signup-to-paid conversion is 19.72%. The subscription summary reports 1,705 active paid customers, $67,895 in current MRR and a 13.54% churn rate.

KPI	Reported result
Users	10,000
Events / Sessions	121,480 / 66,195
Activation	6,314 / 63.14%
Trial	3,097 / 49.05%
Paid	1,972 / 63.67%
Signup-to-paid	19.72%
Active paid customers	1,705
Churned users / churn rate	267 / 13.54%
Current MRR	$67,895

Figures above are transcribed from the project's processed analytical summaries and cross-checked for consistency with those summaries. The SQL scripts were not independently re-executed against a live database for this report.

2. Business Problem & Objectives

SaaS teams need to understand not only how many users sign up, but whether users reach meaningful product value, adopt useful features, convert to paid plans and remain subscribed. ProductScope organizes the available data around these questions.

• Measure acquisition and activation across customer segments.

• Quantify progression through the Signup → Activation → Trial → Paid funnel.

• Compare feature adoption with conversion and churn outcomes.

• Investigate differences by acquisition channel, device and company size.

• Summarize subscription status, plan-level churn and recurring revenue.

• Translate findings into testable product and growth recommendations.

3. Dataset Overview

Four related datasets provide the basis for analysis. The supplied project summaries describe 10,000 users, 121,480 events and 66,195 sessions. Signup dates span January–June 2025, while product event activity extends into September 2025.

Dataset	Analytical role
users.csv	Signup details, acquisition channel, device and company size
events.csv	Product events, timestamps and feature interactions
sessions.csv	Session activity and duration
subscriptions.csv	Trial dates, paid subscription dates, plan, monthly price and cancellation details

Because signup cohorts have different lengths of follow-up, comparisons should use consistent observation windows wherever possible.

4. Analytical Methodology

Data understanding and validation

Python notebooks and SQL checks are used to inspect table structure, record counts, key relationships and date fields. These checks support interpretation of the supplied data but do not guarantee that every business definition is universally applicable.

Funnel analysis

The reported funnel is Signup → Activation → Trial → Paid. Activation is defined in the project SQL as a user having a **project_created** event. Trial and paid stages are identified using **trial_start_date** and **subscription_date**. Counts are distinct users. Unless event ordering is enforced, stage conversion should be interpreted as stage membership rather than proof of strictly sequential progression.

Feature adoption and subscription analysis

Feature adoption covers Dashboard, Collaboration, Export and Integration. The feature adoption output uses activated users as its adoption denominator. Subscription analysis distinguishes paid users, active paid customers and cancelled users. Current MRR is calculated from active subscriptions and monthly price in the supplied analysis.

5. Funnel Performance

Stage	Users	Rate from previous stage
Signup	10,000	100.00%
Activation	6,314	63.14%
Trial	3,097	49.05%
Paid	1,972	63.67%

The largest proportional drop occurs between Activation and Trial: 50.95% of activated users do not appear in the trial stage. This makes the post-activation experience a priority for investigation. Trial-to-paid conversion is 63.67% in the supplied funnel summary.

6. Segment Analysis

Acquisition channel

Channel	Signups	Activation	Trial → Paid
Organic	2,947	64.91%	67.07%
Paid Search	2,516	53.93%	56.95%
Partner	985	67.82%	63.64%
Referral	1,686	71.23%	65.75%
Social	1,866	62.97%	63.75%

Referral has the highest reported activation rate (71.23%), while Paid Search has the lowest (53.93%). Organic contributes the largest active monthly revenue by channel in the supplied output ($20,753), followed by Referral ($15,183). Without acquisition spend and margin data, revenue contribution cannot be interpreted as channel profitability or ROAS.

Device and company size

Desktop activation is reported at 65.69%, compared with 58.48% on Mobile, a 7.21 percentage-point difference. Paid Search on Mobile is the lowest reported channel-device combination at 47.52%. Across company-size groups, activation ranges approximately from 62.50% to 64.73%.

7. Feature Adoption & Outcomes

Feature	Adopters	Adoption*	Trial conv.	Paid conv.	Churn**
Dashboard	3,149	49.87%	53.22%	34.01%	13.82%
Collaboration	2,252	35.67%	51.42%	37.92%	9.02%

Feature	Adopters	Adoption*	Trial conv.	Paid conv.	Churn**
Export	1,689	26.75%	57.43%	36.59%	12.94%
Integration	1,165	18.45%	57.17%	45.41%	7.56%

* Adoption rate uses the activated-user denominator in the feature adoption output. ** Churn is the rate reported for each feature-adopter group; its denominator follows the corresponding processed summary, not all users.

Integration has the lowest adoption rate (18.45%) but the highest reported paid conversion (45.41%) and lowest reported churn (7.56%) among the listed feature groups. This makes Integration discovery and onboarding a promising experiment area. The relationship is observational and does not prove causation.

8. Retention & Subscription Revenue

Plan	Active customers	MRR	Plan churn
Business	188	$18,612	15.32%
Pro	682	$33,418	10.73%
Starter	835	$15,865	15.31%
Total	1,705	$67,895	13.54% overall

Pro contributes the largest MRR ($33,418) and has the lowest reported plan-level churn (10.73%). Starter has the largest active customer count (835). The overall churn figure is 267 cancelled paid users divided by 1,972 paid users (13.54%); it is a cumulative dataset-level rate, not a monthly churn rate without a defined monthly denominator.

MRR represents the sum of monthly prices for active subscriptions in the supplied analysis. It is not recognized revenue or cash collected.

9. Business Recommendations

• **Investigate the activation-to-trial gap.** Map steps after project creation, measure time-to-trial and identify where activated users stop progressing.

• **Review mobile onboarding.** Audit signup completion, project creation, navigation and performance, especially for Paid Search mobile traffic.

• **Test Integration discovery.** Trial guided setup, contextual prompts and use-case education; compare outcomes through a controlled experiment.

• **Evaluate channel quality with cost data.** Combine conversion and revenue with spend, CAC and gross margin before budget decisions.

• **Study Pro-plan retention.** Investigate product capabilities and customer profiles associated with Pro's lower observed churn.

• **Build cohort retention reporting.** Compare signup cohorts at equal elapsed intervals to account for unequal observation windows.

10. Limitations & Interpretation

• The data is observational; associations between feature use, conversion and churn do not establish causality.

• Acquisition cost, campaign spend and gross margin are unavailable, so channel profitability and ROAS cannot be calculated.

• Funnel counts are distinct-user stage counts and may not represent strictly time-ordered progression.

• The 13.54% churn figure is cumulative under the supplied definition, not a monthly churn rate.

• MRR is a subscription run-rate measure, not recognized revenue or cash collection.

• Different signup and event observation windows can affect cohort comparisons.

• Feature-level comparisons may be affected by tenure, user intent and selection effects.

11. Conclusion

ProductScope connects acquisition, activation, product usage, conversion, churn and recurring revenue into a business-oriented view of SaaS performance. The analysis highlights the Activation-to-Trial transition as a key funnel opportunity, weaker activation among Paid Search and Mobile segments, and Integration adoption as a potential area for experimentation. At the subscription level, Pro contributes the largest MRR and shows the lowest observed plan churn.

The next steps are deeper cohort analysis, improved event instrumentation and controlled experiments to validate whether proposed product changes improve activation, conversion and retention.

12. Skills Demonstrated

- Multi-table data understanding and validation
- Python and Pandas for data preparation and analysis
- SQL joins, aggregations and conditional business calculations
- Funnel, segment and feature-adoption analysis
- Subscription churn and MRR analysis
- Power BI KPI reporting and dashboard development
- Translating analytical observations into business recommendations

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GitHub: https://github.com/thegkcode